① $\int_0^1 \frac{x^2}{\sqrt{1-x}}\,dx$ $\qquad \int_0^{\pi/2} \frac{dx}{x^3 \sin x}$

$\int_2^{\infty} \frac{x}{(x^2+3)^{3/2}}\,dx$ $\qquad \int_0^{\infty} \frac{x^2+2}{\sqrt{x^4+1}}\,dx$

"Problemi"? $\underline{x=0}$

$\frac{x^2}{\sqrt{1-x}} \xrightarrow{x \to 1^-} \infty$ di ordine $\frac{1}{2}$

$\boxed{x\to 1^-}$ $\frac{x^2/\sqrt{1-x}}{1/|x-1|^{1/2}} = \frac{x^2}{\sqrt{1-x}} \cdot (1-x)^{1/2} \xrightarrow{x\to 1^-} 1$

Prima studiamo la convergenza, poi i passaggi

$\frac{1}{2} < 1 \Rightarrow \int$ converge!

$\frac{x^2}{\sqrt{1-x}} \sim \frac{1}{|x-1|^{1/2}}$

$x \to 1^-$

$\hookrightarrow$ è finito

$\int_0^1 \frac{x^2}{\sqrt{1-x}}\,dx$ $\qquad \sqrt{1-x} = y$

$\frac{-1}{2\sqrt{1-x}}\,dx = dy$

$1-x = y^2$

$= \int_1^0 (1-y^2)^2 \cdot (-2)\,dy =$

$= 2\int_0^1 (y^4 - 2y^2 + 1)\,dy$

$\hookrightarrow \left[2\frac{y^5}{5} - 4\frac{y^3}{3} + 2y\right]_0^1 = \frac{2}{5} - \frac{4}{3} + 2 = \frac{16}{15}$

$\int_0^{\pi/2} \frac{dx}{x^3 \sin x}$ $\qquad$ "Problemi"

$\underline{x \to 0}$ $\quad \frac{1}{x^3 \sin x} \sim \frac{1}{x^4} \xrightarrow{x\to 0} \infty$

di ordine 4 e quindi

$\sin x \sim x \quad x \to 0$ $\qquad$ l'$\int$ **diverge**:

$\int_0^{\pi/2} \frac{1}{x^3 \sin x}\,dx = +\infty$

$\int_2^{\infty} \frac{x}{(x^2+3)^{3/2}}\,dx$ $\longrightarrow$ tende a 0, ma quanto velocemente?

controllo $x \to \infty$

$\frac{x}{(x^2+3)^{3/2}} \underset{\substack{\uparrow \\ \text{... è asintotico a...}}}{\sim} \frac{x}{(x^2)^{3/2}} = \frac{x}{x^3} = \frac{1}{x^2}$

$\longrightarrow 0$ di ordine $2 > 1$

$\hookrightarrow \int$ converge

$\int_2^{\infty} \frac{x}{(x^2+3)^{3/2}}\,dx$ $\qquad x^2+3 = y \qquad 2x\,dx = dy$

$\int_7^{\infty} \frac{1}{y^{3/2}} \cdot \frac{dy}{2} = \frac{1}{2}\int_7^{\infty} y^{-3/2}\,dy = \frac{1}{2}\left[\frac{y^{-1/2}}{-1/2}\right]_7^{\infty} =$

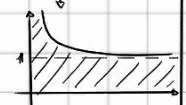
$= -\left[\frac{1}{\sqrt{y}}\right]_7^{\infty} =$

$\left[\lim_{M\to\infty} \frac{1}{2}\left[\frac{y^{-1/2}}{-1/2}\right]_7^{M}\right]$

$= -\left(0 - \frac{1}{\sqrt{7}}\right) = \frac{1}{\sqrt{7}}$

Per $x \to \infty$ $\qquad \frac{x^2+2}{\sqrt{x^4+1}} \sim \frac{x^2}{\sqrt{x^4}} = \frac{x^2}{x^2} = 1$

$\int_0^{\infty} \frac{x^2+2}{\sqrt{x^4+1}}\,dx = \infty$



② $\int_0^{\infty} \frac{|x^2-2x-3| - x^2 - 2x - 3}{x^{5/2}}\,dx$

$x^2 - 2x - 3 \iff (x+1)(x-3) \geq 0$

Per $x \in (0,\infty) \iff x \geq 3$

$= \int_0^3 \frac{-x^2 + \cancel{2x} + \cancel{3} - x^2 - \cancel{2x} - \cancel{3}}{x^{5/2}}\,dx + \int_3^{\infty} \frac{\cancel{x^2} - 2x - 3 - \cancel{x^2} - 2x - 3}{x^{5/2}}\,dx =$

$= \int_0^3 \frac{-2x^2}{x^{5/2}}\,dx + \int_3^{\infty} \frac{-4x-6}{x^{5/2}}\,dx =$

$= -2\underbrace{\int_0^3 x^{-1/2}\,dx}_{} - 2\int_3^{\infty} \frac{2x+3}{x^{5/2}}\,dx$

. Per $x \to 0$ $\quad \frac{1}{\sqrt{x}} \xrightarrow{x\to 0^+} \infty$ di ordine $\frac{1}{2} < 1$

$\hookrightarrow$ OK, converge

. Per $x \to \infty$ $\quad \frac{2x+3}{x^{5/2}} \sim \frac{2x}{x^{5/2}} \sim \frac{2}{x^{3/2}} \xrightarrow{x\to\infty} 0$

di ordine $3/2 > 1$

$\hookrightarrow$ OK, converge

$= -2\left[\frac{x^{1/2}}{1/2}\right]_0^3 - 4\int_3^{\infty} x^{-3/2}\,dx - 6\int_3^{\infty} x^{-5/2}\,dx =$

$= -4\left[x^{1/2}\right]_0^3 - 4\left[\frac{x^{-1/2}}{-1/2}\right]_3^{\infty} - 6\left[\frac{x^{-3/2}}{-3/2}\right]_3^{\infty} =$

$= -4\sqrt{3} + 8\frac{-1}{\sqrt{3}} - 4\frac{1}{3\sqrt{3}} = \frac{-36-24-4}{3\sqrt{3}} = -\frac{64}{3\sqrt{3}}$

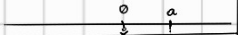
Studio di una funzione integrale

③ Sia $F_a(x) = \int_a^x f(t)\,dt$ $\quad ; f(t) = \frac{1}{t^2 + \sin^2 t}$

$\hookrightarrow$ Ⓘ $\text{dom}(F_a) \to$ come si studia un dominio

$\hookrightarrow$ il dom. di una funzione integrale è un intervallo.

$\text{dom}(f) = (-\infty,0) \cup (0,+\infty)$



Per $t \to 0$: $f(t) \sim \frac{1}{t^2+t^2} = \frac{1}{2t} \xrightarrow{t\to 0} \infty$ di ordine $2 > 1$

$\hookrightarrow$ DIVERGE

Se $a > 0$: $\text{dom}(F_a) = (0,+\infty)$

Se $a < 0$: $\text{dom}(F_a) = (-\infty,0)$

Se $a = 0$: $\text{dom}(F_a) = \{0\}$ ← non c'è niente da studiare

Il dominio di una funzione integrale è un intervallo!!

$\hookrightarrow$ fissiamo $a=1$

$\hookrightarrow$. $\text{dom}(F) = (0,+\infty)$

Ⅱ $\lim_{x\to 0^+} F(x)$ $\qquad \lim_{x\to +\infty} F(x)$

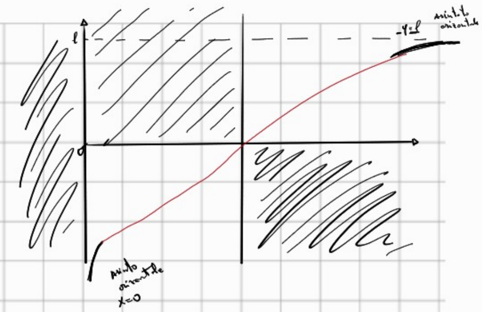
$\lim_{x\to 0^+} F(x) \underset{\substack{\uparrow \\ \text{in senso} \\ \text{IMPROPRIO}}}{``="} \int_1^0 f(t)\,dt$ $\hookrightarrow \infty$ di ord 2 l'$\int$ DIVERGE $\longrightarrow$ f positiva estremi $\int$ scambiati $= \underline{-\infty}$

. $t \to +\infty$ $\quad f(t) = \frac{1}{t^2 + \underset{O(1)}{\sin^2 t}} \sim \frac{1}{t^2} \to 0$ di ord $2 > 1$

$\hookrightarrow \int$ converge

$\lim_{x\to+\infty} F(x) = \int_1^{+\infty} f(t)\,dt = \ell \in \mathbb{R}_+$

↳ sappiamo che ℓ è positivo, non è 0



③ Segno e zeri

$F(x) \geq 0 \overset{f \text{ positiva}}{\iff} x \geq 1$

$F(x) = 0 \iff x = 1$

④ continuità, derivabilità, monotonia di F

F è f.ne integrale $\Rightarrow$ F è continua / derivabile su $(0,+\infty)$

Per $x \in (0,+\infty)$

$F'(x) = f(x) = \frac{1}{x^2 + \sin^2 x} > 0$

F crescente.

$\boxed{4}$ $\quad F_a(x) = \int_a^x \frac{\sqrt{t-1}}{\ln|t^2-2|}\,dt$

$\ln(1+x) \overset{x\to 0}{\sim} x$

. dom F_a $\qquad \lg(y) \xrightarrow{y\to 1} |y-1|$

dom f $\quad t-1 \geq 0 \qquad t \geq 1$

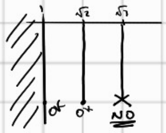
$t^2 - 2 \neq 0 \qquad t \neq \pm\sqrt{2}$

$\ln|t^2-2| \neq 0 \qquad t = \pm\sqrt{3}$

. $\underline{t \to 1^+}$

$f(t) = \frac{\sqrt{t-1}}{\ln|t^2-2|} =$

$\hookrightarrow$ quando $t \to 1$, t^2-2 è negativo



$= \frac{\sqrt{t-1}}{\ln(2-t^2)} = \frac{(t-1)^{1/2}}{\ln(1 + \underline{(1-t^2)})}$ $\quad \ln(1+(1-t^2)) \sim 1-t^2$

$\hookrightarrow x \to \ln(1+x) \sim x$

$\sim \frac{(t-1)^{1/2}}{1-t^2} = \frac{(t-1)^{1/2}}{\underset{\to 2}{(1+t)}(1-t)} \sim \frac{(t-1)^{1/2}}{2(1-t)} = -\frac{1}{2}(t-1)^{1/2}$

$\hookrightarrow$ tende a $-\infty$ di ordine $\boxed{\frac{1}{2} < 1}$

1 è $\underline{\text{OK}}$ $\quad 1 \in \text{dom}F$

$t \to \sqrt{2}$

$f(t) = \frac{\sqrt{t-1}}{\ln|t^2-2|} \longrightarrow 0$

. $t \to \sqrt{3}$

$f(t) = \frac{\overbrace{\sqrt{t-1}}^{\ell \neq 0}}{\ln|t^2-2|} \sim \frac{\sqrt{\sqrt{3}-1}}{\ln(t^2-2)} \sim \frac{\sqrt{\sqrt{3}-1}}{\ln(1+(t^2-3))} \sim$

$\sim \frac{\sqrt{\sqrt{3}-1}}{t^2-3} = \frac{\sqrt{\sqrt{3}-1}}{(t-\sqrt{3})(t+\sqrt{3})} \sim \frac{\sqrt{\sqrt{3}-1}}{\sqrt{3}+3} \cdot \frac{1}{t-3}$

$\hookrightarrow \infty$ di ordine $1 \Rightarrow \int$ DIVERGE

Se $a = \sqrt{3}$ $\quad \{\sqrt{3}\}$

Se $a < 1$: $\emptyset$

Se $1 \leq a$: $[1,\sqrt{3})$

Se $a > \sqrt{3}$: $(\sqrt{3},+\infty)$

⑤ $\int_0^1 \frac{\ln(1+nx) - 6(\sin x)^2}{x^2}\,dx$ $\qquad n \in \mathbb{N}$ $\quad n \geq 0$

$\boxed{x=0?}$

$\frac{1}{x^2} \to \infty$ di ordine 2

$\ln(1+nx) = nx - \frac{n^2}{2}x^2 + \frac{n^3}{3}x^3 + o(x^4)$

$(\sin x)^2 = (x + o(x^3))^2 = x^2 + o(x^4)$

$$\frac{\ln(1+nx) - 6(\sin x)^2}{x^2} = \frac{nx - \frac{n^2}{2}x^2 + \frac{n^3}{3}x^3 - 6x^2 + o(x^4)}{x^2}$$

$\boxed{\underline{\text{Se } n \neq 0}}$: $\sim \frac{nx}{x^2} = \frac{n}{x} \to \infty$ di ordine 1 $\boxed{\Rightarrow \int \underline{\text{DIV}}}$

$\boxed{\text{Se } n = 0}$ $\sim \frac{-6x^2}{x^2} \not\to \infty$ $\boxed{\text{l'}\int\ \underline{\text{converge}}}$

⑥ $\int_0^\infty \frac{\sin x - x^\alpha}{(e^x - 1)^2}\,dx$ $\alpha > 0$

$x \to \infty$: $\frac{\text{Potenza}}{\text{esponenziale}} \to 0$

velocemente $\to$ CONV

$x \to 0$ $\sim \frac{x - \frac{x^3}{6} + o(x^5) - x^\alpha}{x^2}$